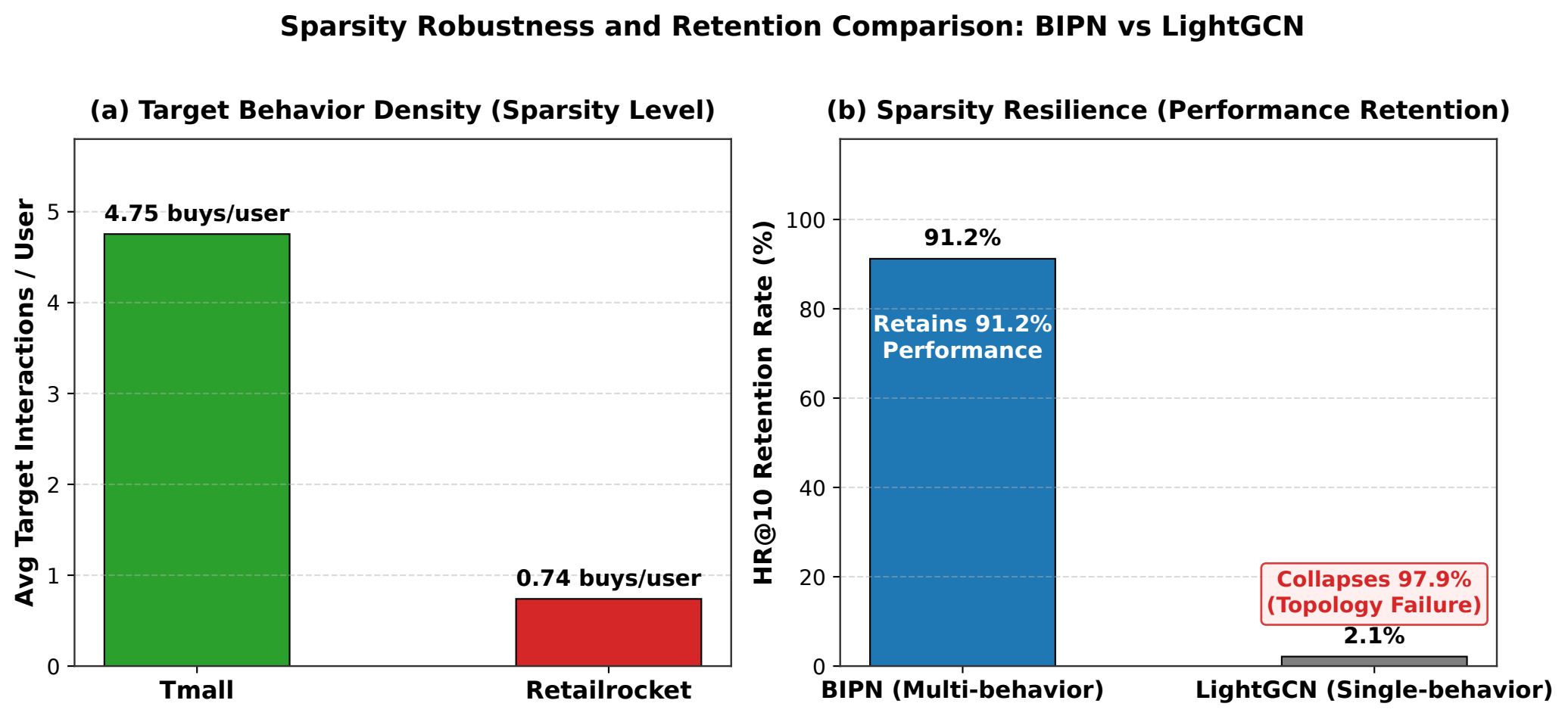
Sparsity Robustness and Retention Comparison: BIPN vs LightGCN
(a) Target Behavior Density (Sparsity Level)
Avg Target Interactions / User
0
1
2
3
4
5
4.75 buys/user
0.74 buys/user
Tmall
Retailrocket
(b) Sparsity Resilience (Performance Retention)
HR@10 Retention Rate (%)
0
20
40
60
80
100
91.2%
Retains 91.2% Performance
Collapses 97.9% (Topology Failure)
2.1%
BIPN (Multi-behavior)
LightGCN (Single-behavior)